Module 5: DOM Manipulation in JavaScript

The **DOM (Document Object Model)** represents an HTML document as a tree of objects. JavaScript uses the DOM to access and modify HTML elements dynamically.

1. Selecting Elements

JavaScript provides several methods to select HTML elements.

getElementById()

Selects an element using its ID.

Example

```
<!DOCTYPE html>
<html>
<body>

<h2 id="title">Welcome Students</h2>

<button onclick="changeText()">
Change Text
</button>

<script>
function changeText() {
document.getElementById("title").innerHTML = "JavaScript
DOM";
}
</script>
```

```
</body>
</html>
```

Output

Welcome Students
↓ Click Button
JavaScript DOM

---

getElementsByClassName()

Selects elements using class name.

```
<p class="demo">HTML</p>
<p class="demo">CSS</p>
<script>
let data =document.getElementsByClassName("demo");
console.log(data[0].innerHTML);
</script>
```

Output

HTML

---

getElementsByTagName()

Selects elements using tag name.

```
<p>Hello</p>
<p>World</p>

<script>
let p =document.getElementsByTagName("p");
```

```
console.log(p.length);
</script>
```

Output

2

---

querySelector()

Returns the first matching element.

```
<p class="text">First</p>
<p class="text">Second</p>

<script>
let data =
document.querySelector(".text");

console.log(data.innerHTML);
</script>
```

Output

First

---

querySelectorAll()

Returns all matching elements.

```
<p class="text">HTML</p>
<p class="text">CSS</p>
```

```
<script>
let data =
document.querySelectorAll(".text");

console.log(data.length);
</script>
```

Output

2

---

2. Events

An Event is an action performed by the user.

Examples:

- Click
- Double Click
- Mouse Over
- Key Press
- Submit

---

onclick Event

```
<button onclick="showMessage()">
Click Me
</button>

<script>
function showMessage() {
```

```
    alert("Button Clicked");
}
</script>
```

---

ondblclick Event

```
<button ondblclick="changeColor()">
Double Click
</button>

<script>
function changeColor() {
    document.body.style.background =
    "yellow";
}
</script>
```

---

onmouseover Event

```
<h2 onmouseover="changeText()">
Move Mouse Here
</h2>

<script>
function changeText() {
    event.target.innerHTML =
    "Mouse Over Detected";
}
</script>
```

---

onkeyup Event

```
<input type="text"
onkeyup="showText(this.value)">

<h2 id="output"></h2>

<script>
function showText(value) {
   document.getElementById("output")
          .innerHTML = value;
}
</script>
```

Practical Example: Live Character Counter

```
<!DOCTYPE html>
<html>
<body>

<textarea id="msg"></textarea>

<h3 id="count">0 Characters</h3>

<script>
document.getElementById("msg")
.addEventListener("input", function(){

   document.getElementById("count")
   .innerHTML =
   this.value.length +
```

```
    " Characters";

});
</script>

</body>
</html>
```

---

3. Forms

Forms collect user input.

Form Validation

Example

```
<!DOCTYPE html>
<html>
<body>

<form onsubmit="return validate()">

<input type="text"
id="username"
placeholder="Username">

<button type="submit">
Submit
</button>
```

```
</form>

<script>
function validate() {

   let user =
   document.getElementById("username")
         .value;

   if(user === "") {

      alert("Username Required");

      return false;
   }

   return true;
}
</script>

</body>
</html>
```

Password Validation

```
<input type="password"
id="password">
<button onclick="checkPassword()">
Check
</button>
<script>
```

```
function checkPassword() {

   let pass =document.getElementById("password").value;
   if(pass.length < 8)
      alert("Minimum 8 Characters");
   else
      alert("Strong Password");
}
</script>
```

Login Form Example

```
<!DOCTYPE html>
<html>
<body>

<input type="text"
id="user"
placeholder="Username">

<input type="password"
id="pass"
placeholder="Password">

<button onclick="login()">
Login
</button>

<h2 id="result"></h2>

<script>
```

```
function login() {

   let user =document.getElementById("user").value;

   let pass =document.getElementById("pass").value;

   if(user === "admin"&& pass === "1234")
  {
      document.getElementById("result")
      .innerHTML =
      "Login Successful";
   }
   else
{
      document.getElementById("result")
      .innerHTML =
      "Invalid Credentials";
 }
}
</script>
</body>
</html>
```

---

4. Dynamic Content

Dynamic Content means changing HTML content without reloading the page.

Change Text Dynamically

```
<h2 id="demo">
Welcome
</h2>

<button onclick="changeText()">
Change
</button>
<script>
function changeText() {
   document.getElementById("demo").innerHTML ="Hello
JavaScript";
}
</script>
```

Change Image Dynamically

```
<!DOCTYPE html>
<html>
<body>

<img id="image" src="https://picsum.photos/200">

<br><br>

<button onclick="changeImage()">
Change Image
</button>

<script>
function changeImage() {
```

```
document.getElementById("image").src =
"https://picsum.photos/200?random=1";

}
</script>

</body>
</html>
```

Add List Items Dynamically

```
<!DOCTYPE html>
<html>
<body>

<ul id="list">
</ul>

<button onclick="addItem()">
Add Item
</button>

<script>
let count = 1;

function addItem() {
let li =document.createElement("li");
li.innerHTML ="Item " + count++;
document.getElementById("list").appendChild(li);
}
</script>
```

```
</body>
</html>
```

Remove Element Dynamically

```
<ul id="list">
<li>HTML</li>
<li>CSS</li>
<li>JavaScript</li>
</ul>

<button onclick="removeItem()">
Remove Last
</button>

<script>
function removeItem() {
let list =document.getElementById("list");
list.removeChild(list.lastElementChild);
}
</script>
```

Practical Project: To-Do List

```
<!DOCTYPE html>
<html>
<body>

<input type="text" id="task">
```

```
<button onclick="addTask()">
Add Task
</button>

<ul id="tasks">
</ul>

<script>
function addTask() {

let task =document.getElementById("task").value;

let li =document.createElement("li");

li.innerHTML = task;

document.getElementById("tasks").appendChild(li);

document.getElementById("task").value = "";

}
</script>

</body>
</html>
```

Real-World Applications

Feature	DOM Concept Used
Login Form	Forms + Events

Feature	DOM Concept Used
OTP Generator	Dynamic Content
Shopping Cart	DOM Manipulation
Dark Mode	Events
Image Slider	Dynamic Content
Live Search	Events
To-Do List	Dynamic Content
Student Registration	Forms
Online Quiz	DOM + Events
E-Commerce Website	All Concepts

Summary

Topic	Purpose
getElementById()	Select by ID
getElementsByClassName()	Select by Class
getElementsByTagName()	Select by Tag
querySelector()	First Matching Element
querySelectorAll()	All Matching Elements
Events	User Actions
Forms	Collect User Input
Dynamic Content	Update Page Without Reload
createElement()	Create New Elements
appendChild()	Add Elements
removeChild()	Remove Elements

Practical DOM Manipulation Examples for Students

These examples are commonly used in real websites and are excellent for lab practicals, viva questions, and mini-projects.

---

1. Show / Hide Password

```
<!DOCTYPE html>
<html>
<body>

<input type="password" id="pass">
<button onclick="togglePassword()">
Show/Hide
</button>

<script>
function togglePassword() {
   let pass = document.getElementById("pass");

   if(pass.type === "password")
      pass.type = "text";
   else
      pass.type = "password";
}
</script>

</body>
</html>
```

---

2. Change Background Color

```
<!DOCTYPE html>
<html>
<body>

<button onclick="changeColor()">
Change Color
</button>

<script>
function changeColor() {

   let colors = [
      "red","blue","green",
      "orange","purple"
   ];

   let random =
   Math.floor(Math.random()*colors.length);

   document.body.style.background =
   colors[random];
}
</script>

</body>
</html>
```

---

3. Font Size Controller

```
<!DOCTYPE html>
<html>
<body>

<p id="text">
JavaScript DOM Manipulation
</p>

<button onclick="increase()">
A+
</button>

<script>
let size = 20;

function increase() {
   size += 2;

   document.getElementById("text")
   .style.fontSize = size + "px";
}
</script>

</body>
</html>
```

---

4. Live Clock

```
<!DOCTYPE html>
<html>
<body>
```

```
<h1 id="clock"></h1>

<script>
function showTime() {

   let now = new Date();

   document.getElementById("clock")
   .innerHTML =
   now.toLocaleTimeString();
}

setInterval(showTime,1000);
</script>

</body>
</html>
```

---

5. Character Counter

```
<!DOCTYPE html>
<html>
<body>

<textarea id="msg"></textarea>

<p id="count">
0 Characters
</p>
```

```
<script>
document.getElementById("msg")
.addEventListener("input",function(){

document.getElementById("count")
.innerHTML =
this.value.length + " Characters";

});
</script>

</body>
</html>
```

---

6. Dynamic Table Generator

```
<!DOCTYPE html>
<html>
<body>

<button onclick="createTable()">
Create Table
</button>

<div id="table"></div>

<script>
function createTable() {

let html =
"<table border='1'>";
```

```
for(let i=1;i<=5;i++) {

html += "<tr>";

for(let j=1;j<=5;j++) {

html += "<td>" +
(i*j) +
"</td>";
}

html += "</tr>";
}

html += "</table>";

document.getElementById("table")
.innerHTML = html;

}
</script>

</body>
</html>
```

---

7. Add Unlimited Notes

```
<!DOCTYPE html>
<html>
<body>
```

```
<input type="text" id="note">

<button onclick="addNote()">
Add Note
</button>

<div id="notes"></div>

<script>
function addNote() {

let text =
document.getElementById("note").value;

let p =
document.createElement("p");

p.innerHTML = text;

document.getElementById("notes")
.appendChild(p);

}
</script>

</body>
</html>
```

---

8. Image Gallery

```
<!DOCTYPE html>
<html>
<body>

<img id="img"
src="https://picsum.photos/300">

<br><br>

<button onclick="nextImage()">
Next
</button>

<script>
function nextImage() {

document.getElementById("img").src =
"https://picsum.photos/300?random="
+ Math.random();

}
</script>

</body>
</html>
```

---

9. Dark / Light Mode

```
<!DOCTYPE html>
<html>
<body>
```

```
<button onclick="toggleTheme()">
Toggle Theme
</button>

<script>
function toggleTheme() {

document.body.classList
.toggle("dark");

}
</script>

<style>
.dark{
background:black;
color:white;
}
</style>

</body>
</html>
```

---

10. Student Percentage Calculator

```
<!DOCTYPE html>
<html>
<body>

<input type="number" id="marks">
```

```
<button onclick="calculate()">
Calculate
</button>

<h2 id="result"></h2>

<script>
function calculate() {

let marks =
Number(
document.getElementById("marks").value
);

let per = (marks/500)*100;

document.getElementById("result")
.innerHTML =
"Percentage = " +
per.toFixed(2) + "%";

}
</script>

</body>
</html>
```

---

11. Dynamic Progress Bar

```
<!DOCTYPE html>
<html>
<body>

<button onclick="start()">
Start Download
</button>

<progress
id="bar"
value="0"
max="100">
</progress>

<script>
function start() {

let value = 0;

let interval =
setInterval(function(){

value += 10;

document.getElementById("bar")
.value = value;

if(value >= 100)
clearInterval(interval);

},500);
```

```
}
</script>

</body>
</html>
```

---

12. Live Search Filter

```
<!DOCTYPE html>
<html>
<body>

<input
type="text"
id="search">

<ul id="list">
<li>Java</li>
<li>Python</li>
<li>JavaScript</li>
<li>PHP</li>
</ul>

<script>
document.getElementById("search")
.addEventListener("keyup",function(){

let value =
this.value.toLowerCase();
```

```
let items =
document.querySelectorAll("#list li");

items.forEach(item => {

item.style.display =
item.innerText
.toLowerCase()
.includes(value)
? "block"
: "none";

});

});
</script>

</body>
</html>
```

---

13. Digital Counter

```
<!DOCTYPE html>
<html>
<body>

<h1 id="count">0</h1>

<button onclick="increase()">
+
</button>
```

```
<script>
let count = 0;

function increase() {

count++;

document.getElementById("count")
.innerHTML = count;

}
</script>

</body>
</html>
```

---

14. Dynamic Card Creator

```
<!DOCTYPE html>
<html>
<body>

<button onclick="createCard()">
Create Card
</button>

<div id="cards"></div>

<script>
function createCard() {
```

```
let card =
document.createElement("div");

card.innerHTML =
"<h2>JavaScript</h2>" +
"<p>DOM Example</p>";

card.style.border =
"1px solid black";

card.style.padding = "10px";

document.getElementById("cards")
.appendChild(card);

}
</script>

</body>
</html>
```

---

15. Mini Shopping Cart

```
<!DOCTYPE html>
<html>
<body>

<button onclick="addItem()">
Add Product ₹500
</button>
```

```
<h2 id="total">
Total: ₹0
</h2>

<script>
let total = 0;

function addItem() {

total += 500;

document.getElementById("total")
.innerHTML =
"Total: ₹" + total;

}
</script>

</body>
</html>
```

Most Important Interview/Lab Questions

1. Create a Password Show/Hide System.
2. Create a Live Character Counter.
3. Create a Dark Mode Toggle.
4. Create an Image Slider using DOM.
5. Create a To-Do List.
6. Create a Student Grade Calculator.
7. Create a Dynamic Table Generator.

8. Create a Shopping Cart Total Calculator.
9. Create a Live Search Box.
10. Create a Digital Clock.
11. Create a Progress Bar.
12. Create a Dynamic Card Generator.
13. Create a Quiz Application.
14. Create an OTP Generator.
15. Create a Color Theme Changer.

These projects demonstrate nearly all important DOM concepts:

- Selecting Elements
- Events
- Forms
- innerHTML
- createElement()
- appendChild()
- removeChild()
- classList
- Dynamic Content Updates
- Event Listeners
- DOM Traversal